

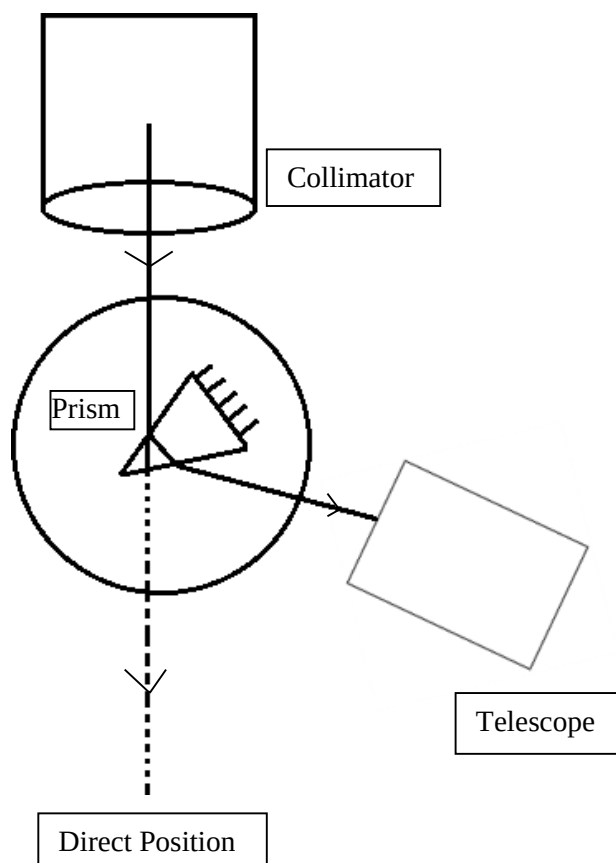
Experiment No. 1

DISPERSIVE POWER OF MATERIAL OF THE PRISM

Aim: To determine dispersive power of the material of the prism.

Apparatus: Spectrometer, Mercury source, prism, table lamp, spirit level

Diagram:



Formula:

$$\omega = \frac{\mu_v - \mu_r}{\left(\frac{\mu_v + \mu_r}{2}\right) - 1}$$

where, ω = dispersive power

μ_v = refractive index for violet color

μ_r = refractive index for red color

Procedure

Preliminary adjustment of the spectrometer

1. Adjust the eye-piece to see the clear image of the straight wire.
2. The telescope is directed towards a distant object and the distance between the eyepiece and the objective lens is adjusted to get a clear image. The telescope now set for parallel rays.
3. A spirit level is placed on the prism table and the screws at the base are adjusted till it is perfectly horizontal.
4. The telescope is brought in line with the collimator and the focus knob on the collimator is adjusted, if required. The direct reading R_0 is noted.
5. Fix the Vernier table and mount the prism on the turn table to observe the spectrum.
6. Turn the telescope to observe the red line.
7. Now rotate the table so that, the image moves towards the axis of collimator, i.e angle of deviation decreases.
8. The rotation is continued till the image stops and begins to retrace its path. This gives position of the minimum deviation.
9. The vertical cross wire is made to coincide with the image and note the reading R_o .
10. Repeat the above procedure to obtain the angle of minimum deviation for violet line and note the corresponding reading R_v .
11. The difference between the direct reading and refracted reading for the colors gives angle of minimum deviation (d) for corresponding spectral lines.
12. Use the given formula to calculate the dispersive power of the prism.

Observation:

$$\text{Least Count (L.C.)} = \frac{\text{Value of 1MSD (in degree)}}{\text{No. of divisions on V.S.}} = \dots\dots\dots$$

$$\text{L.C.} = \text{-----degree} = \text{-----minute}$$

Direct reading			Line	Spectrometer reading			Angle of min. deviation $\delta = R - R_0 $ degree
MSR degree	VSD	TR degree		MSR degree	VSD	TR Degree	
			Red				
			Violet				

MSR = Main Scale Reading

VSD= Coinciding Vernier Scale Division

TR = Total Reading = MSR + (VSD $\times$ L. C.) (e.g. 118.5 + 20*(1/60) = 118.83)

Calculations:

1

$$\mu_r = \frac{\sin\left(\frac{A + \delta_r}{2}\right)}{\sin\frac{A}{2}}$$

, (Where A is the angle of Prism, A = 60°)

2

$$\mu_v = \frac{\sin\left(\frac{A + \delta_v}{2}\right)}{\sin\frac{A}{2}}$$

, (Where A is the angle of Prism, A = 60°)

3. Dispersive power of the prism calculated by following formula

$$\omega = \frac{\mu_v - \mu_r}{\left(\frac{\mu_v + \mu_r}{2}\right) - 1}$$

Results:

Dispersive power of the material of the given prism is

Precaution:

1. Take the readings without any parallax errors
2. The focus should be at the edge of Red and violet rays
3. The slit should be as narrow as possible but the knife edges of the slit should not touch each other.
4. The telescope and the collimator should be separately set for parallel rays.
5. The height of the prism table should be so adjusted that the maximum light must fall on the entire surface
Of the prism
6. While taking observations the telescope and the prism table must be clamped.